

The Chinese Remainder Theorem

Discrete Mathematics
Courtesy of Andrei Bulatov

Previous Lecture

● Congruence modulo m

- $a \equiv b \pmod{m}$ if $m \mid (a - b)$

● Residues modulo m

- Remainder $0 \leq r < m$ when dividing by m

● Integers modulo m

- $\mathbb{Z}_m = \{0, 1, \dots, m-1\}$

● Modular arithmetic

- Addition, subtraction, and multiplication as operations on $\mathbb{Z}_m$

+	0	1	2	3	4
0	0	1	2	3	4
1	1	2	3	4	0
2	2	3	4	0	1
3	3	4	0	1	2
4	4	0	1	2	3

.	0	1	2	3	4
0	0	0	0	0	0
1	0	1	2	3	4
2	0	2	4	1	3
3	0	3	1	4	2
4	0	4	3	2	1

Divisors of Zero

- *Modular addition just rotates/shifts/cycles the numbers*
 - *In particular, it is **injective** as a function $f(x) = x + a \pmod{m}$*
- *Multiplication however is **not** injective over $\mathbb{Z}_4$*

.	0	1	2	3
0				
1				
2				
3				

- *A **proper 0 divisor** modulo m is a residue $a \not\equiv 0 \pmod{m}$ such that there exists $b \not\equiv 0 \pmod{m}$ with $a \cdot b \equiv 0 \pmod{m}$.*
- *$\mathbb{Z}_4$ has a proper 0 divisor. $\mathbb{Z}_5$ does not.*

Inverse

● A residue b modulo m is called an **inverse** of a residue a if $a \cdot b \equiv 1 \pmod{m}$, denoted a^{-1}

● What is...

- The inverse of 2 modulo 5?
- The inverse of 2 modulo 4?

● Theorem

Let a be residue modulo m . The following are equivalent:

- (i) a has an inverse;
- (ii) a is not a proper divisor of 0;
- (iii) a is **relatively prime** with m .

Inverse (cntd)

● Proof.

(i) $\Rightarrow$ (ii) *By contradiction.*

Suppose $a \cdot b \equiv 0 \pmod{m}$ for some b not equal to 0.

Then $a^{-1} \cdot a \cdot b \equiv a^{-1} \cdot 0 \pmod{m}$

$b \equiv 1 \cdot b \equiv 0 \pmod{m}$, a contradiction

(ii) $\Rightarrow$ (iii) *By contraposition.*

Suppose $\gcd(a, m) = d$ and $a = ld$, $m = kd$.

Then $ak \equiv kld \equiv lm \equiv 0 \pmod{m}$.

Thus a is a proper divisor of 0.

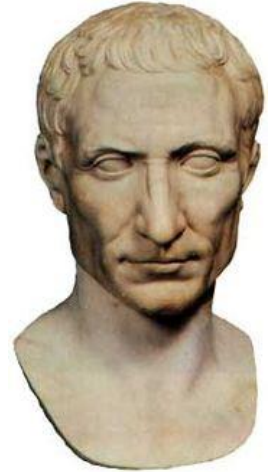
(iii) $\Rightarrow$ (i)

Suppose $\gcd(a, m) = 1$. Then there are u, v with $au + mv = 1$.

Thus $au \equiv 1 \pmod{m}$; a has an inverse.

Applications: Cryptography

- *One of the oldest cryptosystems is the Caesar cipher. He made messages secret by shifting each letter three letters forward. Thus B becomes E, and X is sent to A*
- *To express this process mathematically we first replace letters by integers from 0 to 25. For example, A is replaced by 0, K by 10.*
- *Next, to encrypt a message we add 3 modulo 26 to every letter.*
- *Finally, replace numbers with corresponding letters*
- *To decrypt a message, perform all the actions above in the reverse order*



Applications: Cryptography (cntd)

● *Encrypt 'SEND MORE MEN AND MUNITION'*

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

S E N D M O R E M E N A N D M U N I T I O N Z
 18 4 13 3 12 14 17 4 12 4 13 0 13 3 12 20 13 8 19 8 14 13 25

21 7 16 6 15 17 20 7 15 7 16 3 16 6 15 23 16 11 22 11 17 16 2
V H Q G P R U H P H Q D Q G P X Q L W L R Q C

Question: can the receiver ever decrypt a different message than the one that was sent? I.e. is decryption unambiguous?

Applications: Cryptography (cntd)

- Caesar cipher with a key. A key is just a word, e.g. 'KEY'
- Replace it with numbers: 10 4 24
- Then the message is encrypted by adding 10 to the first letter, 4 to the second letter, 24 to the third letter, 10 to the fourth letter and so on.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
S	E	N	D	M	O	R	E	M	E	N	A	N	D	M	U	N	I	T	I	O	N	Z			
18	4	13	3	12	14	17	4	12	4	13	0	13	3	12	20	13	8	19	8	14	13	25			
10	4	24																							
2	8	11	13	16	12	1	8	10	8	11	10	17	1	22	24	11	18	23	6	24	17	23			
C	I	L	N	Q	M	B	I	K	I	L	K	R	B	W	Y	L	S	X	G	Y	R	X			

Pseudorandom Generators

- *Randomly chosen numbers are often needed for computer simulations. However, truly random numbers are very difficult to obtain.*
- *This is why people mostly use **pseudorandom** numbers*
- *The most commonly used procedure for generating pseudorandom numbers is the **linear congruential** method*
- *We choose four numbers: the **modulus** m , **multiplier** a , **increment** c , and **seed** x_0 , with $2 \leq a < m$, $0 \leq c < m$, $0 \leq x_0 < m$*
- *We generate a sequence of pseudorandom numbers $\{x_n\}$, with $0 \leq x_n < m$ for all n , by repeatedly using the congruence*
$$x_{n+1} \equiv ax_n + c \pmod{m}$$
- *Typical values: $m = 2^{32}$, $a = 1664525$, $c = 1013904223$*

Linear Congruences

- A congruence of the form

$$ax \equiv b \pmod{m}$$

where m is a positive integer, a and b are integers, and x is a variable, is called a **linear congruence**.

- We often need to **solve** linear congruences for x

- If a is relatively prime with m , then it has the inverse a^{-1} . Then

$$a^{-1} \cdot ax \equiv a^{-1} \cdot b \pmod{m}$$

$$x \equiv a^{-1} \cdot b \pmod{m}$$

- Example:

Solve the linear congruence $3x \equiv 4 \pmod{7}$

Systems of congruences

- Sometimes an unknown x is defined by multiple linear congruences, called a **system of congruences**

- (Sun Tzu's puzzle, 400 – 460 BC):

“There are certain things whose number is unknown. When divided by 3, the remainder is 2; when divided by 5, the remainder is 3; and when divided by 7, the remainder is 2. What will be the number of things?”

- This can be translated into the following question: What are the solutions of (i.e. values of x satisfying) the system of congruences

$$x \equiv 2 \pmod{3}$$

$$x \equiv 3 \pmod{5}$$

$$x \equiv 2 \pmod{7}$$

The Chinese Remainder Theorem

● Theorem

Let $m_1, m_2, \dots, m_k$ be pairwise relatively prime positive integers
and $a_1, a_2, \dots, a_k$ arbitrary integers. Then the system

$$\begin{aligned} x &\equiv a_1 \pmod{m_1} \\ x &\equiv a_2 \pmod{m_2} \\ &\vdots \\ x &\equiv a_k \pmod{m_k} \end{aligned}$$

has a unique solution modulo $m = m_1 \cdot m_2 \cdot \dots \cdot m_k$.

That is, there is a solution x with $0 \leq x < m$, and all other solutions are congruent modulo m to this solution.

Alternatively, the solutions have the form $c + dm$

The Chinese Remainder Theorem (cntd)

● *Proof (Existence)*

Set $M_i = \frac{m}{m_i}$ for $i = 1, 2, \dots, k$ --- i.e. $M_i = m_1 m_2 \dots m_{i-1} m_{i+1} \dots m_k$

Since m_i and m_j are relatively prime when $i \neq j$, $\gcd(M_i, m_i) = 1$

So, M_i has an inverse $M_i^{-1} \pmod{m_i}$, i.e. $M_i M_i^{-1} \equiv 1 \pmod{m_i}$

Now set $x = a_1 M_1 M_1^{-1} + a_2 M_2 M_2^{-1} + \dots + a_k M_k M_k^{-1}$.

Since $M_j \equiv 0 \pmod{m_i}$ when $i \neq j$, we see that x is a solution:

$$\begin{aligned} x &= a_1 M_1 M_1^{-1} + a_2 M_2 M_2^{-1} + \dots + a_k M_k M_k^{-1} \equiv a_i M_i M_i^{-1} \pmod{m_i} \\ &\equiv a_i \pmod{m_i} \end{aligned}$$

The Chinese Remainder Theorem (cntd)

● *Proof (Uniqueness)*

Assume x and y are solutions to the system of congruences

Then $x \equiv y \pmod{m_i}$ for each i

But this means that $(x-y)$ is a multiple of m_i for every m_i

Hence $(x-y)$ is a multiple of $m = \text{LCM}(m_1, \dots, m_k) = m_1 m_2 \dots m_k$

Hence $x \equiv y \pmod{m}$

Sun Tzu's Puzzle

$$x \equiv 2 \pmod{3}$$

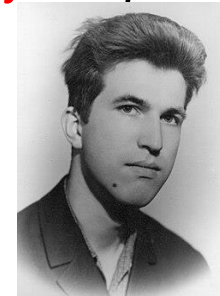
$$x \equiv 3 \pmod{5}$$

$$x \equiv 2 \pmod{7}$$

Diophantine equations

- *Systems of linear congruences are examples of a larger class of equations or massive importance in mathematics: **Diophantine equations***
- *Diophantine equations are equations of the form*
$$ax^n + by^m + cz^p + \dots = d$$
where $x, y, z, \dots$ are unknown integers to be solved for
- *Linear congruences are Diophantine equations of the form*
$$x = y + mz$$
- *Hilbert's tenth problem*

*Devise an algorithm which can solve **any** Diophantine equation*
- *Hilbert's tenth problem was closed in 1970 by Yuri Matiyasevich, who showed **no such algorithm can exist***



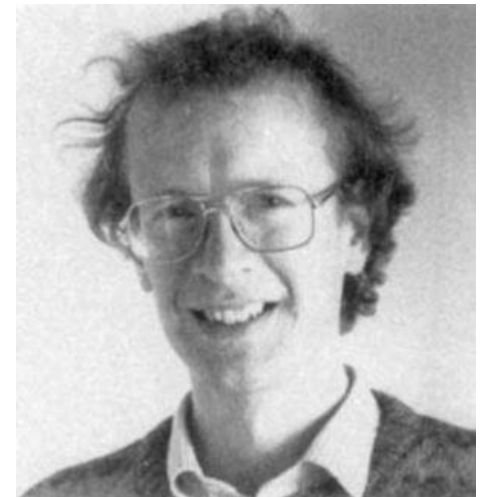
Fermat's Theorem

- **Fermat's Great (Last) Theorem.**

For any $n > 2$, the equation $x^n + y^n = z^n$ does not have integer solutions $x, y, z > 0$

- *It had remained unproven for 358 years (posed in 1637, proved in 1995)*

- *Andrew Wiles proved it in 1995 with a 100 page proof using group theory, Galois theory, and tools from algebraic geometry*



Fermat's Little Theorem

● Fermat's Little Theorem.

If p is prime and a is an integer not divisible by p , then

$$a^{p-1} \equiv 1 \pmod{p}$$

● *Clearly, it suffices to consider only residues modulo p .*

$\mathbb{Z}_5$

.	0	1	2	3	4
0	0	0	0	0	0
1	0	1	2	3	4
2	0	2	4	1	3
3	0	3	1	4	2
4	0	4	3	2	1

Fermat's Little Theorem (cntd)

- *Fermat's Little Theorem was improved by Euler*

- **Fermat's Little Theorem improved**

For any integers m and a such that they are relatively prime

$$a^{\varphi(m)} \equiv 1 \pmod{m}$$

where $\varphi(m)$ denotes the Euler totient function, the number of numbers $0 < k < m$ relatively prime with m

- *Example: $\mathbb{Z}_8$*

Practice

Exercises from the Book:

7th edition: 17, 20, 31, 33, 35 (page 284 – 285)

8th edition: 17, 20, 31, 33, 35 (page 301)